

Chemotherapy in Gynaecology

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Chemotherapy / Hormone Therapy in Gynaecology

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Outline

Chemotherapy: definition & tumour growth · Mechanisms of drug action · Pharmacology & toxicity · Remission · Drug resistance · Principles of clinical chemotherapy · Chemotherapy in gynaecological cancers · Gestational trophoblastic disease · Side effects · Hormone therapy

Chemotherapy – definition and tumour growth

Chemotherapy is the treatment of disease using **chemical substances**. The **ideal agent selectively kills tumour cells with minimal toxicity to normal tissues**.

Tumour growth: cells progress through the **cell cycle** to divide; the **generation time** is the time to complete one cycle.

Tumour cells do NOT have a faster generation time than normal cells. However, tumours have **more cells in the active phases of replication**, so they generally grow faster. Chemotherapy exploits this difference to preferentially kill tumour cells.

The **doubling time** is the time for a tumour mass to double in size; it varies widely, and **metastases generally have faster doubling times than the primary lesion**.

Mechanisms of drug action

Class	Examples	Mechanism
Alkylating agents	Melphalan, ifosfamide, cyclophosphamide	Cross-link DNA strands , preventing protein synthesis and cell division
Antimetabolites	Methotrexate, 5-fluorouracil	Inhibit metabolic processes required for DNA/RNA synthesis
Anti-tumour antibiotics	Dactinomycin, mitomycin, bleomycin, adriamycin (doxorubicin)	Natural products from soil fungi
Plant alkaloids	Etoposide, vinblastine, vincristine, taxol	Arrest cells in metaphase by binding microtubular protein of the mitotic spindle
Miscellaneous	Cisplatin, carboplatin, hexamethylmelamine	Do not fit the above categories

Pharmacological considerations & toxicity

- Chemotherapy works by **first-order kinetics** – killing a **constant FRACTION** of cells, not a constant number
- Most agents are given **intravenously**, and eliminated by the **kidneys** or metabolised in the **liver**
- **Drug toxicity** = unwanted effects from injury to normal cells; it **limits the dose**
- Agents are used at doses producing **some toxicity** to be maximally effective
- Toxicity is quantified by the **therapeutic index** (ratio of therapeutic dose to toxic dose)
- **Haematologic toxicity** (leukopenia, thrombocytopenia, anaemia) is the **most frequent**; also **GI** (mucositis, diarrhoea) and **skin** (alopecia)

Clinical remission

- **Complete remission** – disappearance of **all** objective evidence of tumour and complete resolution of all signs and symptoms
- **Partial remission** – a reduction of **at least 50%** in all measurable lesions, some subjective improvement, and **no new lesions** during therapy

Drug resistance

A major problem. By the **Goldie-Coldman hypothesis**, most cells start **sensitive** but develop **resistance at variable rates**.

Tumours are only curable if no resistant cell lines are present or develop. To minimise resistance, **multiple-drug regimens are preferred to single-drug therapy**.

Pleiotropic (multidrug) resistance – certain resistance mechanisms confer **cross-resistance to structurally dissimilar drugs** with different mechanisms.

Principles of clinical chemotherapy

Before starting:

- **Histological classification** of the tumour is essential
- Determine the **therapeutic index** relative to the **stage**
- Evaluate the role of chemotherapy/hormones vs other modalities
- Assess **patient factors** – age, performance status, organ reserve, comorbidity
- **Start with the most effective agent** (cheaper long-term, avoids prolonged hospitalisation)
- Ensure **infrastructure** – trained personnel and sterile mixing facilities

Summary of successful drug therapy: an **effective agent** at the **best dose/schedule**; a tumour with a **high growth fraction**, **accessible** to drugs (not in a sanctuary), with **small volume or large fractional kill** (to avoid resistant clones); and **normal tissue** that recovers faster than the tumour regenerates.

Chemotherapy in gynaecological cancers

Ovarian cancer – responds **best** of all gynaecological cancers; chemotherapy is **standard after initial surgery**. First-line: **platinum (cisplatin/carboplatin) + paclitaxel**. If hypersensitive to paclitaxel: cyclophosphamide or topotecan. **Intraperitoneal chemotherapy** delivers sustained high doses to the peritoneum while sparing marrow. **Malignant germ-cell tumours:** platinum-based **BEP (bleomycin, etoposide, cisplatin)**.

Endometrial cancer – chemotherapy mainly for **advanced/metastatic/recurrent** disease. Most active: **doxorubicin, platinum, paclitaxel**. **Medroxyprogesterone acetate (MPA)** 200-400 mg/day gives ~40% response but is not valuable as an adjuvant.

Cervical cancer (squamous) – increasingly shown sensitive to cytostatics. **Chemoradiation** (chemotherapy + concurrent radiotherapy) is now an important modality. Most effective agents: **cisplatin, or cisplatin + 5-FU**. Platinum-based chemoradiation gives a **~30% reduction in risk of death** in locally advanced disease vs radiotherapy alone.

Vaginal cancer – chemotherapy (including cisplatin) has **no significant activity**.

Vulval cancer – used in **advanced, unresectable** disease: **mitomycin C + 5-FU** with external beam radiotherapy.

Gestational trophoblastic disease (GTD)

Indications for chemotherapy:

- **-hCG > 20,000 IU/ml persisting 4 weeks** after evacuation
- **Rising -hCG**
- **Histological choriocarcinoma**
- **Metastasis**

Patients are categorised by the **WHO scoring system** (age, antecedent pregnancy, -hCG level, interval, tumour size, site and number of metastases, prior chemotherapy – each scored 0-4):

Risk	Score	Regimen
Low	0-4	Methotrexate alone ($\pm$ folinic acid) or actinomycin D

Risk	Score	Regimen
Medium	5-7	Combination – EMA-CO (etoposide, methotrexate, actinomycin D, cyclophosphamide, vincristine)
High	> 8	EMA-CO ; if it fails, cisplatin/etoposide/bleomycin/ifosfamide

Side effects of chemotherapy (drug-specific)

- **Nausea & vomiting** – metoclopramide, dexamethasone, lorazepam, 5-HT antagonists
- **Bone marrow depression** – almost all agents
- **Phlebitis/sclerosant** – especially **adriamycin**
- **Neurologic** – bleomycin, vincristine
- **Skin exfoliation** – bleomycin
- **Pulmonary** – bleomycin, methotrexate
- **Nephrotoxicity** – cisplatin (less with carboplatin)
- **Hepatotoxicity** – methotrexate
- **Cardiotoxicity** – adriamycin (doxorubicin)
- **GIT** – methotrexate, 5-FU, cyclophosphamide
- **Alopecia** – almost all
- **Stomatitis** – methotrexate, 5-FU, adriamycin

Hormone therapy

Hormones are chemical substances produced by one organ that act on distant **target organs**. Indications in gynaecology:

- **Hormonal contraceptives** – Oral (COCs, progestin-only); injectables (DMPA, Noristerat); implants (Norplant, Uniplant, LNG-IUS); **post-coital (Yuzpe regimen** – 2 pills of 50 µg ethinylestradiol + 0.5 mg norgestrel within 72 h, repeated 12 h later); IUD insertion or mifepristone
- **Hormone replacement therapy (HRT)** – vasomotor symptoms (hot flushes; **oestradiol** mainly); **osteoporosis** (the most important menopausal hazard – lowest effective dose, e.g. 0.625 mg conjugated oestradiol); genitourinary atrophy; **hypogonadism** (sexual maturation, epiphyseal closure); premature ovarian failure; hypothalamic-pituitary suppression (exercise-induced amenorrhoea, anorexia nervosa)
- **Infertility / assisted reproduction, pregnancy maintenance, abnormal uterine bleeding, hyperandrogenism/hirsutism**, gynae-oncology/endometriosis, and miscellaneous (e.g. urethral mucosal prolapse)

References

Standard O&G oncology texts; lecture material of Prof E O Ezirim.